Enabling power meter reading applications

MG301-M Series Modular MASS Modem

· LTE CAT 1

· MASS Protocol

· IEC62056-21

· DLMS/COSEM

· Li-ion Battery

1. Product Overview

MG301 is a wireless modem designed according to the requirements of the Automatic Meter Reading System (OSOS) for communication via cellular networks, and MG301-M is the modular model that will operate with modular meter. MG301-M enables the server to access the connected metering devices via the cellular network, allowing the server to obtain the operational parameters of the meters at any time and record detailed traceability information. MG301-M is equipped with various hardware interfaces such as serial port, digital input, relay output, etc., enabling it to connect to various devices on-site and achieve multiple applications.

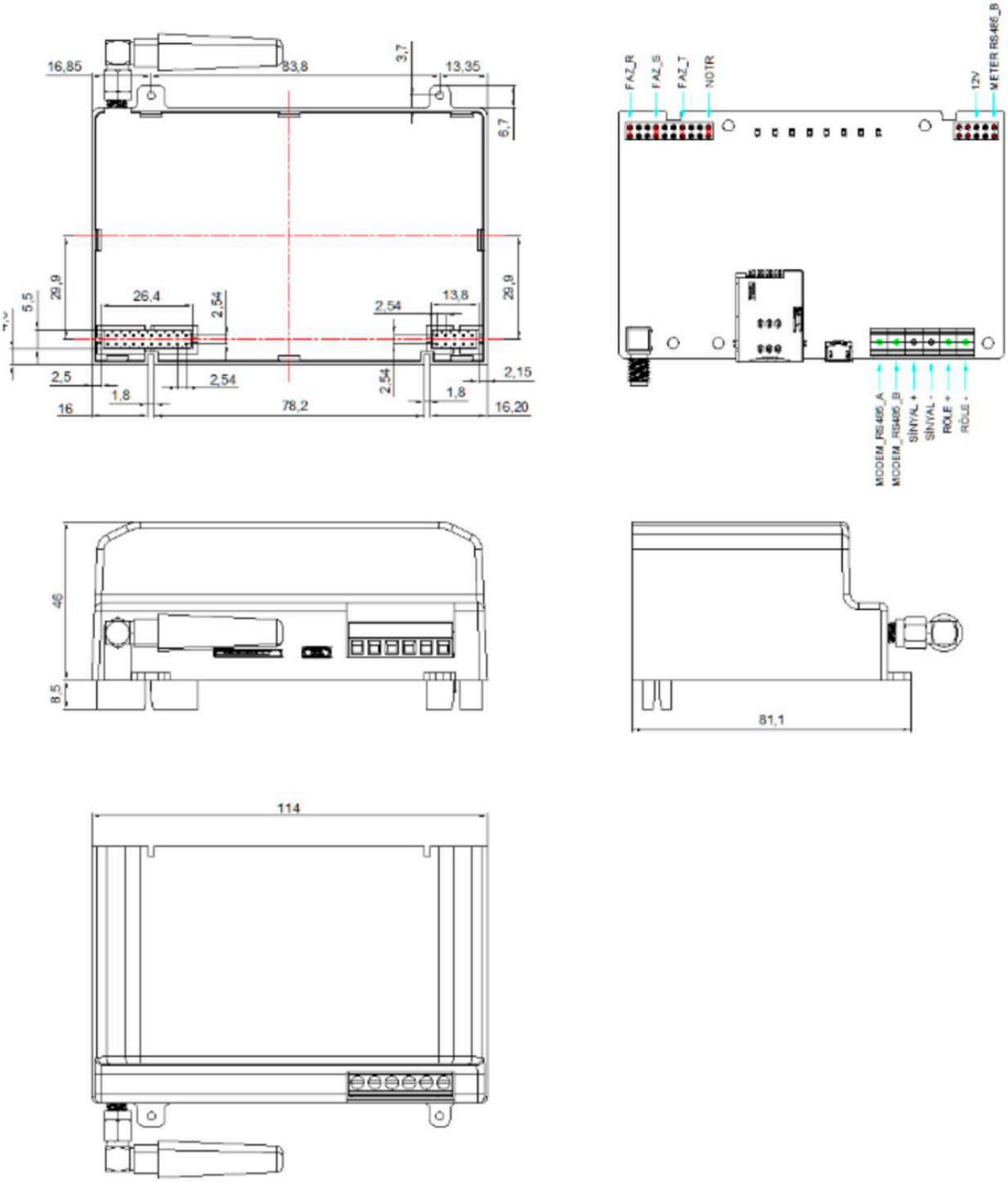
Key Technical Specifications

Specification	Details
Cellular	2G/4G (LTE CAT 1)
Hardware Platform	2.5MB RAM; 32MB Flash
Metering Protocols	IEC62056-21 Mode-C; DLMS/COSEM
Platform Access	MQTT/MQTTS/VIKO/MASS/TeleMetriX Head-End
Interfaces	2 × RS485; 1 × USB2.0 Type-C; 1 × digital input; 1 × relay output
Power	12V DC ± 10%; 6W (MAX)
Dimensions	114 × 74.4 × 45 mm
Operating Environment	-30 to 70 °C; 0–93% ambient humidity
Protection	IP40

2. Features and Advantages

Feature	Description
Performance and Platform	2.5MB RAM; 32MB Flash
Rich Interfaces	2 × RS-485; 1 × USB2.0; 1 × digital input; 1 × relay output
Stable Communication	2G/4G (LTE CAT 1); independent hardware and software watchdog; multilevel link detection, communication self-healing
Industrial-grade, Ready for Harsh Industrial Environments	Operating temperatures up to -30 ~ 70 °C; protection rating IP40; Li-ion battery
Multiple Acquisition Protocol	IEC62056-21; DLMS/COSEM; MASS; MQTT/MQTTS; VIKO; TeleMetriX Head-End

3. Product Dimensions



Dimension (specification table): 114 × 74.4 × 45 mm.

4. Product Interfaces

Interface	Description
Cellular	2G/4G (LTE CAT 1)
USB	1 × USB2.0, Type-C
Serial Port	2 × RS485; RS485: Data+, Data-, GND
SIM Card	Mini SIM-2FF card, supports 1.8/3V SIM/UIM card
Antenna Connector	1 × Cellular Antenna
Button	1 × System reset button
Digital Input	1 × digital input; dry contact: Logic 0, Open; Logic 1, Close
Relay Output	1 × relay output: 240VAC/5A, latching relay

LED Indicators

LED	Status Description
PWR	Steady on: The power supply is normal Off: No external power supply is available
IP	Steady on: The device obtains an IP address for network communication Off: The device does not obtain an IP address for network communication
RS485-1	Blinking: Data is being exchanged through serial port 1 Off: No data is being exchanged over serial port 1
RS485-2	Blinking: Data is being exchanged through serial port 2 Off: No data is being exchanged over serial port 2

LED	Status Description
Relay	Steady on: Closing Off: Open
2G/4G	Blinking: 2G Steady on: 4G Off: No Communication
3 × CSQ	Signal quality indicator

5. Hardware Specifications

Category / Parameter	MG301-M
Hardware Platform	
RAM	2.5MB
FLASH	32MB
Interfaces	
Cellular	2G/4G (LTE CAT 1)
USB	1 × USB2.0, Type-C
Serial Port	2 × RS485; RS485: Data+, Data-, GND
SIM Card	Mini SIM-2FF card, supports 1.8/3V SIM/UIM card
Antenna Connector	1 × Cellular Antenna
Button	1 × System reset button
Digital I/O	1 × digital input: Dry contact: Logic 0, Open; Logic 1, Close; 1 × relay output: 240VAC/5A, Latching Relay
Other Hardware	

Category / Parameter	MG301-M
RTC	Real time clock operated from on-board coin-cell battery
Punctuality Accuracy	0.5S / day
Battery	Li-ion Battery, 300mAh, rechargeable and replaceable.
Power	
Power Supply	12V DC ± 10%
Power Consumption	6W (MAX)
Mechanical Characteristics	
Mounting	Plug to the modem socket of the modular meters
Enclosure Material	Plastic
Dimension	114 × 74.4 × 45 mm
Protection	IP40
Ambient Temperature and Humidity	
Operation Temperature	-30 ~ 70℃
Ambient Humidity	0–93%
Declared Degree of Contamination	3
Overvoltage Category	CAT III
Electrical Protection Class	II

Category / Parameter	MG301-M
LED	
System	1 × Power, 3 × CSQ, 1 × IP, 1 × 2G/4G, 2 × RS485, 1 × Relay
Safety & EMC	
Safety & EMC	IEC 62368-1; EN 55032, Class B; EN 55035, Level 2
Other Standards	
Others	EN 60068-2-30; EN 60068-2-31; EN 60068-2-1; EN 60068-2-2; EN 62262; IEC 60529; IEC 60695-11-10 standard V-0 class

6. Software Specifications

Parameter	Specification
Operating System	
OS	RTOS
Internet	
Access Authentication	APN, VPDN
Internet Access	CHAP, PAP
Network Format	2G/4G (LTE CAT 1)
eSIM	Support eSIM (LPA)

Parameter	Specification
Network Protocol	
IP Application	ICMP, DNS, TCP/UDP, TCP Server, DHCP, NTP
Reliability	
Power Failure Alarm	Detecting an external power failure
Link Detection	Sends heartbeat packets to detect, auto redials when disconnected
Embedded Watchdog	Device runs self-detection, auto recovers from malfunctions
Temperature Detection	The CPU temperature can be detected and recorded in logs
Communication Protocol	
Acquisition Protocol	IEC62056-21 Mode-C; DLMS/COSEM
Access Platform	MQTT/MQTTS/VIKO/MASS/TeleMetriX Head-End
Security	
Encryption	AES-128 and ECDHE

Parameter	Specification
Authentication	In cases where configuration changes, software updates, or connection to the control center are made via the modem's serial ports, authentication will be performed between the modem and the device attempting to connect to it. A log will be kept for these operations.
Network Management	
Configuration	Web Via USB
Upgrade	Local Via USB, Remote Via FTP/InHand DM platform (Coming soon)
Log	Support local system log and important log power-off autosave

7. Ordering Information

Model Code

Model: MG301-M-FQ53

Product Models

Model	Cellular Type & Frequency Band	Serial Port	USB	I/O
MG301-M-FQ53	EMEA CAT1 FDD: B1/B3/B7/B8/B20/B28 TDD: B38/B40/B41 GSM: B2/B3/B5/B8	2 × RS485	1 × USB2.0	1 × Digital Input 1 × Relay Output

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